

Two component system

Reduced phase rule or condensed phase rule.

We know the phase-rule equation, $F = C - P + 2$ (1)

For a two component system, $C = 2$ and hence the above equation becomes,

$$F = 2 - P + 2 = 4 - P \text{ (2)}$$

The minimum number of phases in any system at equilibrium is one. It is clear from the equation (2), the maximum number of degree of freedom is three.

Thus, three variables – pressure, temperature and composition of one of the components must be specified to describe the system. This will lead to three dimensional figures which cannot be conveniently represented on a paper. To make this simple, one of the three variables is kept constant.

In solid-liquid equilibrium of an alloy, practically there is no gaseous phase and the pressure will not have much influence. In the case of solid-liquid equilibrium, the experiments are generally carried out at constant pressure.

Thus the system in which only the solid and liquid phases are considered and the gas phase is ignored is called a condensed system. This reduces the degree of freedom of the system by one. The phase rule equation is then written as

$$F' = C - P + 1 \text{ (3)}$$

This equation is called reduced phase rule or condensed phase rule.

For a two component system the phase rule equation is written as

$$F' = C - P + 1 = 2 - P + 1 = 3 - P \text{ (4)}$$

The above equation is known as the reduced (condensed) form of phase rule for two component system.

Classification of two component system

The two component systems are classified into the following three types :

- i) Simple eutectic formation
- ii) a) Formation of compound with congruent melting point. b) Formation of compound with incongruent melting point.
- iii) iii) Formation of solid solution.

i) Simple eutectic formation :

A system with two substances which are completely miscible in the liquid state, but completely immiscible in the solid state is known as eutectic system. In this system the substances do not react chemically. Among the mixtures of different proportions of two substances, the mixture which has the lowest melting point is known as the eutectic mixture. The temperature and composition corresponding to the point eutectic point is called eutectic temperature and eutectic composition respectively.

ii) a) Formation of compound with congruent melting point : In this type of binary alloy system the two substances form one or more compounds with definite proportions. Of the compounds, a compound is said to possess congruent melting point, if it melts exactly at a constant temperature into liquid, having the same composition as that of the solid.

b) Formation of compound with incongruent melting point : Of the above compounds, a compound is said to possess incongruent melting point, if it decomposes completely at a temperature below its melting point yielding a new solid phase with a composition different from that of the original.

iii) Formation of solid solution : In this type, when two substances, especially metals, are completely miscible in both the solid and liquid states, they form solid solutions where

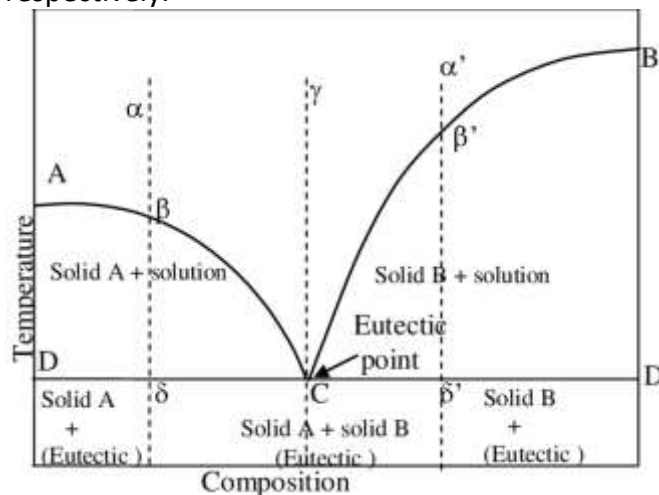
mixing takes place in the atomic levels. A solid solution can be formed only when the difference between the atomic radius of two metals is not greater than 15%.

Simple eutectic systems

The general phase diagram for binary alloy systems is shown in Fig 4.2 . Here the pressure does not have the considerable effect. Hence, the other two variables viz, temperature and compositions are taken into account.

Components A and B.

When small quantities of B are added to A gradually, the melting point of A falls along the curve AC. In the same way when small quantities of A are added to B gradually, the melting point B falls along the curve BC. Hence, AC and BC are the freezing point curves of A and B respectively.



The Simple eutectic systems

The curves AC and BC meet at the point C. At this point the three phases solid A, solid B and their solution coexist. The degree of freedom is zero here and the system is therefore invariant. Also only at this point C, the liquid can exist at the lowest temperature. Since the mixture of components A and B of composition corresponding to the point C has the lowest melting point, the point C is called the eutectic point. The temperature and composition corresponding to the point C is called eutectic temperature and eutectic composition respectively.

Consider a liquid mixture of composition represented by a point α cooled at constant pressure. The temperature falls without any change of composition until the point on the curve AC is reached. At this temperature t_1 , the solid A will separate out. The system now consists of two phases and hence monovariant. As cooling continues, the component A keeps on separating out and the solution becomes relatively richer in B. The temperature and the solution composition both change along AC. Thus at the temperature t_1 , solid A is in equilibrium with solution of composition X and at temperature t_2 , it is in equilibrium with solution of composition Y.

It is clear therefore, in the area ACD, solid A is in equilibrium with solutions of varying composition given by the curve AC depending upon the temperature.

When the temperature reaches a point represented by δ , the solid B also begins to separate out. On further cooling the system, solid A and B separate out together in constant ratio so that the composition of the solution remains constant. The temperature also remains constant for some time. When the liquid solution has been completely solidified

and the system consists only of a mixture of solid A and B, it becomes monovariant. Further cooling will result in the fall of temperature below the line DD into the area in which only the two solids coexist as shown.

In the same way, if the composition of liquid mixture is on the right of the eutectic point C, as represented by point α' , similar series of changes will be obtained on cooling.

Merits of the Phase rule:

1. It is applicable to both physical and chemical equilibria.
2. It requires no information regarding molecular/micro-structure, since it is applicable to macroscopic systems.
3. It is a convenient method of classifying equilibrium states in terms of phases, components and degrees of freedom.
4. It helps us to predict the behaviour of a system, under different sets of variables.
5. It indicates that different systems with same degree of freedom behave similarly.
6. It helps in deciding whether under a given set of conditions :
 - a) various substances would exist together in equilibrium or
 - b) some of the substances present would be interconverted or
 - c) some of the substances present would be eliminated.

Limitations of Phase rule:

1. It can be applied only for system in equilibrium. Consequently, it is of little value in case of very slow equilibrium state attaining system.
2. It applies only to a single equilibrium system; and provide no information regarding any other possible equilibria in the system.
3. It requires at most care in deciding the number of phases existing in an equilibrium state, since it considers only the number of phases, rather than their amounts. Thus even if a trace of phase is present, it accounts towards the total number of phases.
4. It conditions that all phases of the system must be present simultaneously under the identical conditions of temperature and pressure.
5. It conditions that solid and liquid phases must not be in finely-divided state; otherwise deviations occurs.

ENGINEERING MATERIALS

POLYMERS

Polymers are natural and synthetic macromolecules formed by the repeated union of several simple molecules called monomers. A monomer is a simple molecule having two or more functional groups or bonding sites through which each can link to the other monomers to form a polymer chain. The properties of polymers are entirely different from those of its monomers. The fundamental chemical process by which a large number of small molecules of low molecular weight combine to form macromolecules of high molecular weight is called polymerization.

Classification of polymers

Polymers are classified in different ways based on their sources, thermal behavior, mechanism of polymerization and properties as given below:

- a) Natural and synthetic polymers
- b) Thermoplastics and thermosetting polymers
- c) Addition and condensation polymers
- d) Elastomers, fibers, resins and plastics
- e) Homo polymers and copolymers

a) Natural and synthetic polymers:

This type of classification is based on the source of polymers. The polymers which are obtained from natural sources such as plants and animals are called natural polymers. Eg. Cotton, wool, silk, starch, cellulose, proteins, natural rubber, etc.

The polymers which are synthesized from simple molecules are called synthetic polymers. eg. Polyethylene, PVC, Teflon, Nylon, etc.

b) Thermoplastics and thermosetting polymers

The basis for this classification is the thermal behavior of the polymers. Thermoplastics are the polymers which become soft on heating and hard on cooling. This cycle can be repeated many times without affecting their chemical properties. Eg. Polyethylene, PVC, Teflon, etc.

Thermosetting polymers are polymers which undergo chemical changes on heating and become permanently hard, rigid and infusible. They will not soften on heating, once they are set. Eg. Phenol- formaldehyde resin, urea- formaldehyde resin, etc.

c) Addition and condensation polymers

The basis for this classification is their polymerization mechanism.

The polymers which are formed by the addition polymerization reaction in which self addition of several olefin monomers to each other takes place without elimination of by products are called addition polymers. Eg. PVC, polyethylene, polystyrene, etc.

The polymers which are formed by the intermolecular condensation reactions through functional groups of monomers with continuous elimination of by products are called condensation polymers. Eg. Nylons, polyesters, urea-formaldehyde, etc.

d) Elastomers, fibers, resins and plastics

Polymers can also be classified based on their structure and end uses.

i) Elastomers: The polymer which gets elongated when pulled, but returns to the original length on release of force are called an elastomer. Eg. Natural rubber, Buna-S, etc.

ii) Fibers: Fibers are long, thin and thread-like polymer chains which do not undergo stretching or deformation like elastomers. Eg. Wood, silk, cotton, nylons, terylene, etc

iii) Plastics: The polymers which can be molded into desired articles by the application of heat and pressure are called plastics. Eg. PVC, Teflon, etc.

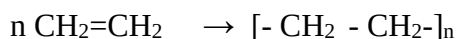
iv) Resins: Low- molecular mass polymers are called resins. Eg. Phenol-formaldehyde resin, urea – formaldehyde resin, etc.

e) Homo polymers and copolymers

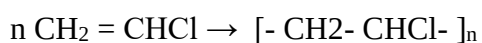
Polymers formed from the same type of monomers are known as homo polymers. Eg. Polythene, PVC, nylon6, etc. if two or more different monomers make up the polymer, it is known as copolymer. Eg. Nylon66, styrene-butadiene rubber, etc.

General methods of production of polymers

1. By addition polymerization: A polymerization reaction in which monomers containing one or more double bonds are linked to each other without elimination of any byproduct is called addition polymerization. It is also known as chain polymerization since it is a chain reaction initiated by a radical or an ion. For example, ethylene polymerizes into polyethylene at 100-200°C and 100-300 atm. Pressure and in presence of benzoyl peroxide.



Ethylene Polyethylene



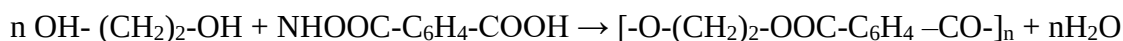
Vinyl chloride PVC

Only olefinic compounds can undergo addition polymerization. The elemental composition of the polymer is same as that of the monomer. The molecular weight of the polymer is an integral multiple of that of the monomer.

2. By condensation polymerization: A polymerization reaction in which bi- or poly functional monomers undergo intermolecular condensation with continuous elimination of byproducts such as H_2O , HCl , NH_3 , etc is called condensation polymerization. The reaction is normally catalyzed by acids or bases. Some examples are as follows;



Hexamethylene diamine Adipic acid Nylon6,6



Ethylene glycol Terephthalic acid Terylene (polyester)

The monomers having two or more reactive functional groups can undergo condensation polymerization. The elemental composition of the polymer is generally different from that of its monomers.

Mechanism of polymerization

Depending on the mechanism involved, addition polymerization can be of 4 different types:

(i) Free radical polymerization, (ii) cationic polymerization, (iii) anionic polymerization and (iv) coordination polymerization

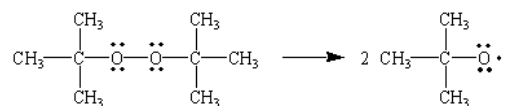
1.Free radical polymerization

Addition polymerization that takes place by free radical mechanism involves three distinct stages: initiation, propagation and termination

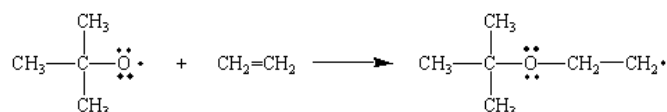
A. Initiation

This process involves two reactions.

(i) This part involves the production of free radicals by the homolytic dissociation of an initiator, such as di-tert-butyl peroxide to yield a pair of radicals, either by heat or light. A free radical is an atomic or molecular species having an odd or unpaired electron. They are highly active species.

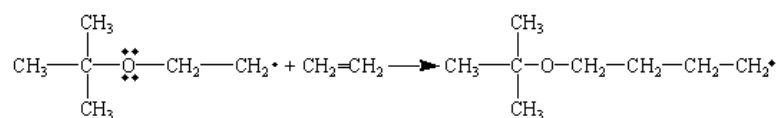


(ii). This part of initiation involves the addition of this radical to the first monomer molecule to produce the chain initiating species.



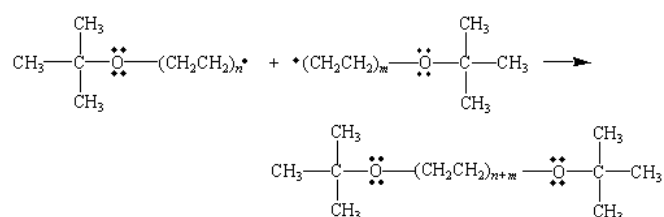
B. Propagation

In propagation, the radical attacks another monomer to produce yet another free radical and the process continues until termination occurs.



C. Termination

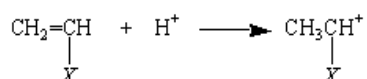
Whenever pairs of radicals combine to form a covalent bond, the chain reactions carried by these radicals are terminated.



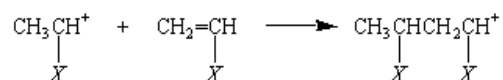
2. Cationic polymerization

In cationic polymerization, the polymerization is initiated by a cation obtained either from protonic acids (HCl, HNO₃, H₂SO₄) or Lewis acids (AlCl₃, BF₃, etc.). Some common examples of cationic polymerization are homopolymerization of isobutylene and copolymerization of isobutylene with 2-methyl propene. The various steps involved in the mechanism are explained as follows:

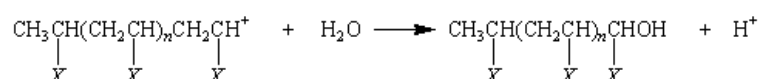
(i) Initiation: The addition of a cation to the double bond of the monomer leads to the formation of a carbonium ion and this initiates the polymerization.



(ii) Propagation: In this step, the carbonium ion formed adds to the double bond of another monomer, generating a bigger carbonium ion and the chain propagates in a similar manner.



(iii) Termination: The chain reaction is terminated when the carbonium ion reacts with traces of water in the solvent in which the reaction is run.

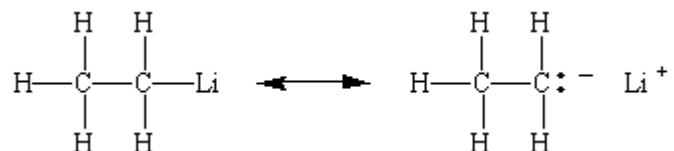


3. Anionic polymerization

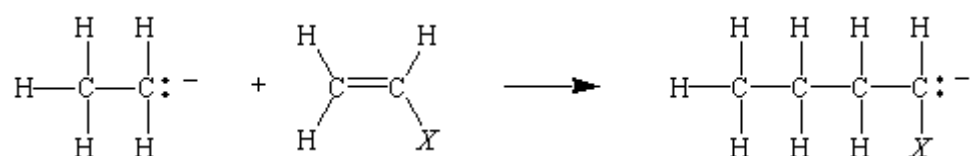
When the chain reaction is initiated and carried by negatively charged intermediates, the reaction is known as **anionic polymerization**. Like free-radical polymerizations, these

chain reactions take place via chain-initiation, chain-propagation, and chain-termination steps.

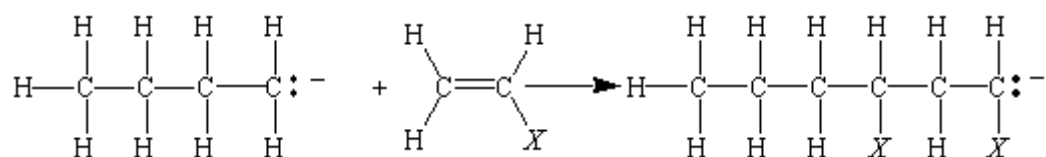
The reaction is initiated by a Grignard reagent or alkyllithium reagent, which can be thought of a source of a negatively charged CH_3^- or CH_3CH_2^- ion.



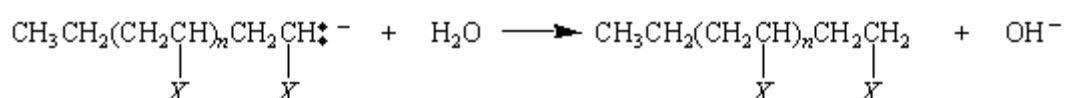
The CH_3^- or CH_3CH_2^- ion from one of these metal alkyls can attack an alkene to form a carbon-carbon bond.



Where X is an electron withdrawing group. The product of this chain-initiation reaction is a new carbanion that can attack another alkene in a chain-propagation step. The presence of an electron withdrawing group facilitates the reaction by stabilizing the carbanion formed.



The chain reaction is terminated when the carbanion reacts with traces of water in the solvent in which the reaction is run.

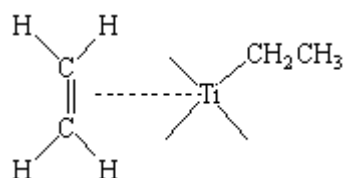


Some common examples of anion polymerization include polymerization of styrene, acrylonitrile and methyl methacrylate.

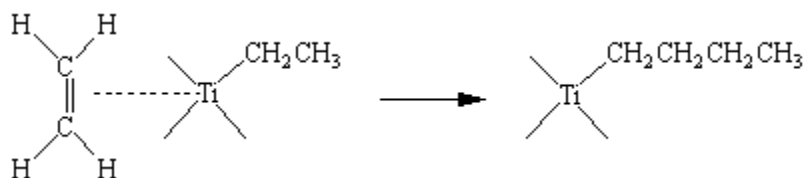
Coordination polymerization

In coordination addition polymerization, an organometallic active intermediate acts as the initiator. This generally involves the use of Ziegler-Natta catalysts, which is a metal-organic complex.

A typical Ziegler-Natta catalyst can be produced by mixing solutions of titanium(IV) chloride (TiCl_4) and triethylaluminum [$\text{Al}(\text{CH}_2\text{CH}_3)_3$] dissolved in a hydrocarbon solvent. An alkene acts as a Lewis base toward the titanium atom, donating a pair of electrons to form a transition-metal complex.



The alkene is then inserted into a Ti-CH₂CH₃ bond to form a growing polymer chain and a site at which another alkene can bond.



Ziegler-Natta catalysts produce a more linear polymer, which is more rigid, with a higher density and a higher tensile strength. Polypropylene produced by free-radical reactions, for example, is a soft, rubbery, atactic polymer with no commercial value. Ziegler-Natta catalysts provide an isotactic polypropylene, which is harder, tougher and more crystalline.

Plastics

Plastics are polymers of either natural or synthetic organic compounds with high molecular mass. They may contain added substances to improve their properties and performance. The word plastic is derived from the Greek word 'plastikos', which means capable of being shaped or molded. It refers to the property called plasticity which allows the material to be cast, pressed or extruded into variety of shapes. Plastics are used in the manufacture of a variety of industrial products such as films, fibers, plates, tubes, bottles, boxes, etc.

Classification of plastics

1. Thermoplastic polymers: These are plastics which become soft on heating and hard on cooling. The process of heating and cooling can be repeated for a number of times without affecting the properties. Eg. Polythene, PVC, teflon, etc
2. Thermosetting polymers: These are plastics which undergo chemical changes and cross linking on heating and become permanently hard , rigid and infusible on cooling. Eg. Phenol-formaldehyde, urea-formaldehyde, etc.

Difference between thermoplastics and thermosetting plastics

Thermoplastics	thermosetting plastics
1.They are mostly formed by addition polymerization	They are formed by condensation Polymerization
2.They exhibit linear long chain structures With negligible cross links	They have three dimensional network structure joined by strong covalent bonds
3.They undergo reversible changes On heating	They undergo irreversible changes on heating, and degrade
4. They consist of polymers of low molecular weight	They consist of polymers of high molecular weight
5.They are soft, weak and less brittle	They are strong, hard and more brittle
6. They are soluble in organic solvents	They are insoluble in organic solvents
7. They readily soften on heating	They do not, soften on heating, but causes Degradation on prolonged heating
8. By heating to suitable temperatures, They can be softened, reshaped and	They retain their shape and structure even on heating. Hence, they cannot be reshaped and

reused

reused

9. There is no change in chemical

They undergo chemical changes during molding

Composition and structure during

process

Molding process

Compounding of plastics

The majority of the commercial plastic products are not made of only pure polymers. Before fabricating commercial plastic products, they are generally mixed with various additives to enhance their mechanical and aesthetic properties. Such process is called compounding.

The major types of compounding ingredients or additives are:

Fillers or extenders: These are inert materials added to polymers to enhance their mechanical strength and also to reduce the cost of the final plastic product. Particulate fillers such as asbestos powder, sawdust, mica, talc, clay and fibrous fillers such as cotton, pulp, fiberglass, etc, are used.

Plasticizers: These additives increase the plasticity and flexibility of the polymer. Phthalate esters, oils, camphor, tricresyl phosphate, tributyl phosphate are the commonly used plasticizers. Plasticizer molecules penetrate into the polymer and reduce the intermolecular forces of attraction between polymer chains. This increases the mobility of the polymer chains as they slide over each other. Thus, plasticizers act as an internal lubricant.

Stabilizers: Most polymers undergo thermal, chemical or photolytic degradation during processing of plastics. Stabilizers are additives that tend to chemically stabilize the polymer from degradation. Organometallic salts, lead chromate, alkaline earth oxides, and epoxy compounds are used as stabilizing agents. Due to the negative impact of the use of heavy metals, lead stabilizers are usually replaced with calcium and zinc salts.

Resins: They act as binding agents. Natural or synthetic resins are added to the polymer to hold all the additives together.

Catalysts: Various antioxidants such as peroxides, ammonia and certain transition metals are added to accelerate the formation of cross links in thermoset polymers.

Lubricants: They provide a glossy finish to the final commercial plastic product thereby enhancing the aesthetic appeal. Lubricant additives such as waxes and oils also prevent plastics from sticking to the molding equipment.

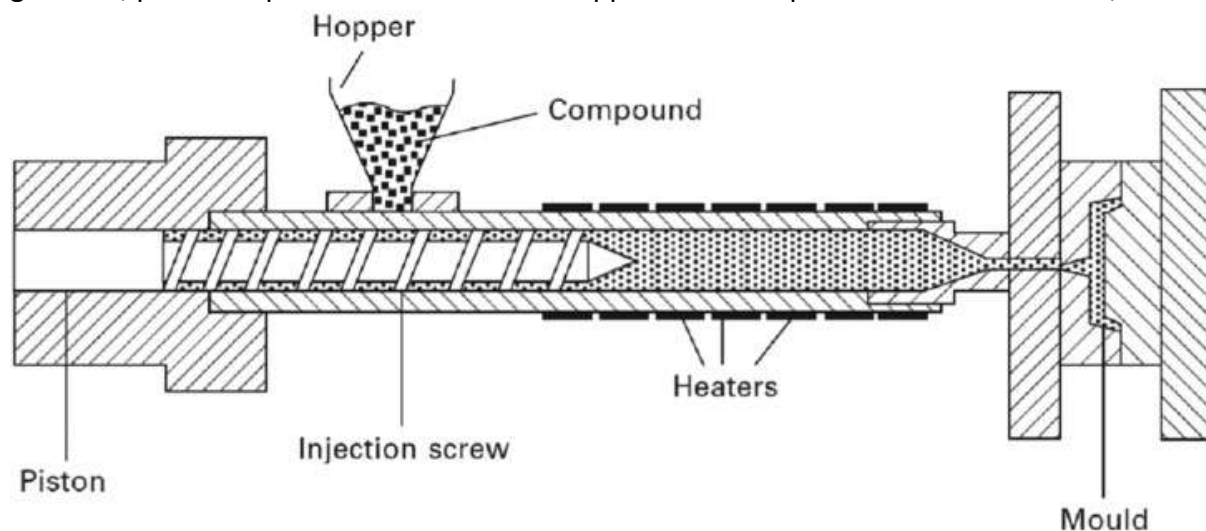
Coloring materials: Organic dyes and pigments impart the desired color. Carbon black, ultramarine (blue), zinc oxide(white), ferric oxide(red), etc. are some commonly employed coloring pigments.

Molding techniques of plastics

Plastic molding involves the conversion of the solid polymeric material into a desirable size and shape. During molding, the plastic material is heated to the appropriate temperature for it to flow and the material is shaped and then cooled to preserve the desired shape. For fabricating thermosetting plastics, molten mix should be molded as quickly as possible before it cools down because once molding mix attains curing temperature, it sets permanently. Continuous molding methods such as injection molding and extrusion methods are not suitable for thermosetting polymers, which can be molded by compression or transfer molding techniques.

Injection molding

In this type of molding process, a known quantity of polymer which may be in the form of granules, pellets or powder is fed into the hopper. It is then passed to a hot chamber,



where the polymer material softens and flows under the pressure applied by means of an electrically operated plunger. (Piston in the form of a rotating screw). As the plunger moves in the forward direction, the softened material passes into the mold through a nozzle, solidifies upon cooling and takes the shape of the mold. Then mold is opened and the article is removed.

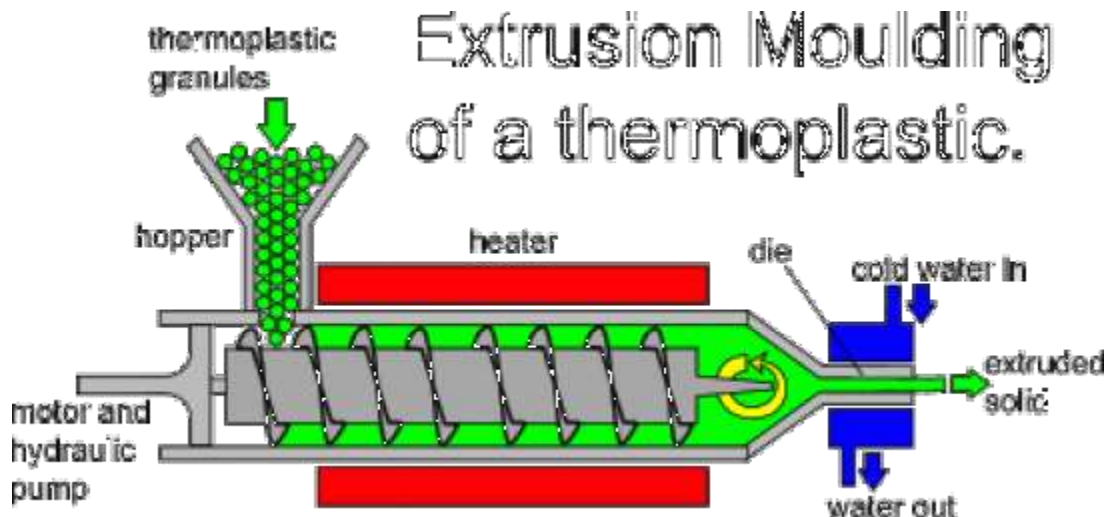
Advantages:

1. High speed production
2. Low costs in mass production
3. Very low loss of materials.
4. Flexibility to make parts with complex shapes
5. High precision

Extrusion molding

This technique is a type of injection molding used for the continuous production of thermoplastic material of uniform cross section such as rods, tubes, sheets and cables.

As with injection molding, dry plastic material is placed into a hopper and fed into a long heating chamber.



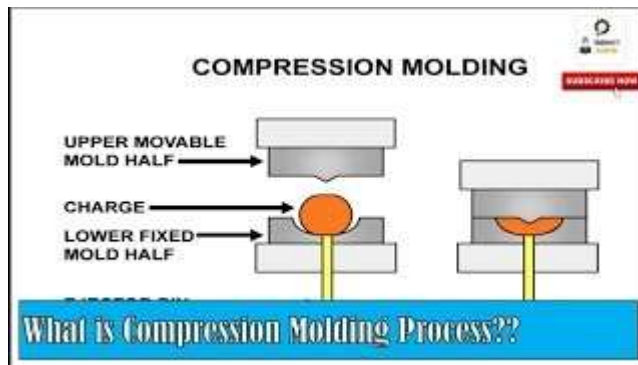
At the end of the chamber, however, the material is forced out of a small opening or a die in the shape of the desired finished product. As the plastic exits the die, it is placed on a conveyor belt where it is allowed to cool. Blowers are sometimes used to aid in this process, or the product may be immersed in water to help it cool.

Advantages:

1. Low initial setup costs
2. Fast setup time
3. Low production cost

Compression molding

Compression molding is the most common process used with thermosetting materials and is usually not used for thermoplastics.



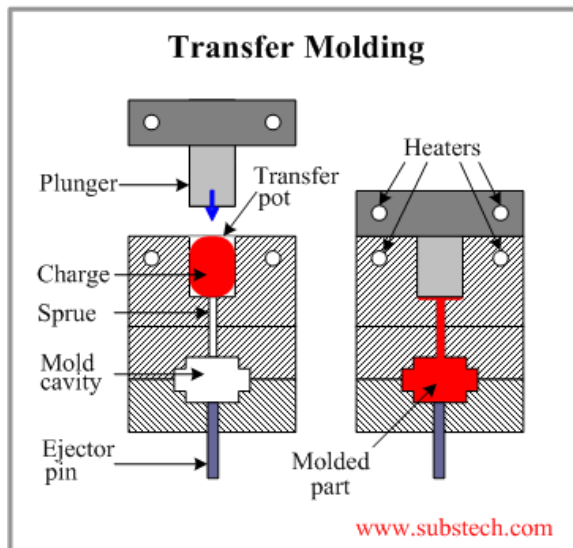
A known quantity of polymer is placed in the lower portion of the mold cavity. It is heated electrically to soften the material. The mold cavity gets filled with the fluidized plastic. The melt is compressed to about $100\text{--}500\text{kg/cm}^2$ by the upper mold. The compressed mass gets cured and hardened and is opened after cooling and the article is removed.

Advantages:

1. Low initial setup cost.
2. Very low loss of material.
3. Good surface finish
4. Fast setup time

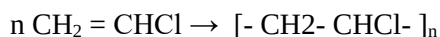
Transfer molding

It is a combined compression- injection process in which the raw material is plasticized by heat and pressure in a separate chamber. In this process, a thermosetting polymer is loaded into a chamber, immediately ahead of the mold cavity, where it is heated. Pressure is then applied to force the softened polymer into the heated mold where curing occurs and finally the molded part is ejected.



Poly vinyl chloride (PVC)

PVC is produced by free-radical initiated chain polymerization of vinyl chloride. Vinyl chloride is subjected to highly reactive free-radical initiators such as benzoyl peroxide in the temperature range 50-75°C. Under the action of the initiators, the double bond in the vinyl chloride monomers is opened, and one of the resultant single bonds is used to link together thousands of vinyl chloride monomers to form the repeating units of polymers.



Vinyl chloride PVC

Properties: 1) PVC is the most versatile of all plastics because of its blending capability with plasticizers, stabilizers and other additives.

2) It is resistant to light, atmospheric oxygen, inorganic acids and alkalis

3) It possesses greater stiffness and rigidity compared to poly ethylene.

4) It is a non-flammable, and chemically inert polymer.

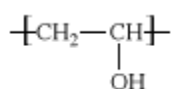
Applications: PVC is used in an enormous range of domestic and industrial products, from raincoats and shower curtains to window frames and indoor plumbing. Approximately 60% of PVC is used in building and construction, mainly in rigid pipes and roof coatings. Plasticized PVC is used mainly in wire and cable insulations and in packaging. Other important applications are pipe fittings, electrical outlet boxes and parts for automotive bumpers.

Polyvinyl alcohol (PVA)

It is a colourless, water-soluble synthetic resin employed principally in the treating of textiles and paper.

PVA is unique among polymers in that it is not built up in polymerization reactions from monomers. Instead, PVA is made by dissolving another polymer, polyvinyl acetate (PVAc), in an alcohol such as methanol and treating it with an alkaline catalyst such as sodium hydroxide. The resulting

hydrolysis, or “alcoholysis,” reaction removes the acetate groups from the PVAc molecules without disrupting their long-chain structure. The chemical structure of the resulting vinyl alcohol repeating units is

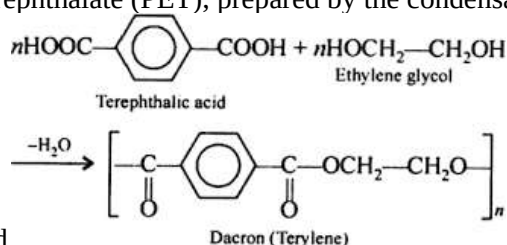


When the reaction is allowed to proceed to completion, the product is highly soluble in [water](#) and insoluble in practically all organic solvents. Incomplete removal of the acetate groups yields resins less soluble in water and more soluble in certain organic liquids. Polyvinyl alcohol has excellent film forming, [emulsifying](#) and [adhesive](#) properties. It is also resistant to oil, grease and [solvents](#). It has high tensile strength and flexibility.

PVA is used in [sizing](#) agents that give greater strength to textile yarns and make paper more resistant to oils and greases. It is also employed as a component of adhesives and emulsifiers, as a water-soluble protective film, and as a starting material for the preparation of other resins. By reaction with butyraldehyde ($\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$) and [formaldehyde](#) (CH_2O), PVA can be made into the resins [polyvinyl butyral](#) (PVB) and polyvinyl formal (PVF). PVB, a tough, clear, adhesive, and water-resistant [plastic](#) film, is widely used in laminated [safety glass](#), primarily for automobiles. [PVF](#) is used in wire insulation.

Terylene or polyethylene terephthalate (PET)

These are prepared by the condensation polymerization of diols with dibasic acids. Terylene or polyethylene terephthalate (PET), prepared by the condensation between ethylene glycol and



terephthalic acid

Properties: 1. PET can exist both in amorphous and (transparent) and semi crystalline (white) form.

2. It has high mechanical strength and is stable over temperature range of - 40 to 100°C.

3. It has abrasion resistance and good electrical insulation properties.

4. At room temperature it is resistant to water, dilute acids, salts and alcohols.

Applications: 1. Used for making clear bottles for foods and beverages

2. Used in electrical appliances as insulators.

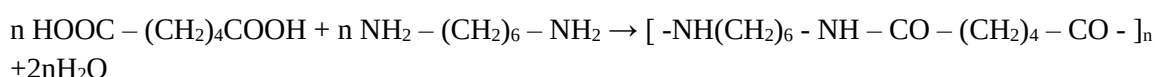
3. Used to make fabrics, upholstery, etc.

4. Terylene fiber can be blended with wool and other natural fibers.

Nylon

Nylons are condensation polymers that belong to the class of polyamides. Nylons are condensation polymers that belong to the class of polyamides. Two of the ingredients that are used to synthesize the most common nylon, adipic acid and hexamethylenediamine, each contain six carbon atoms, and the product has been named nylon-6,6.

Nylon6,6 is prepared by the condensation copolymerization of adipic acid with hexamethylene diamine in equimolar ratio.

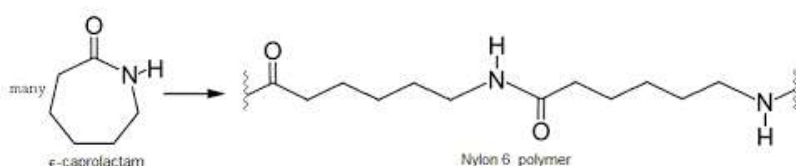


Adipic acid

Hexamethylene diamine

Nylon-6, 6

When caprolactam is the starting material, nylon-6 is obtained, so named because it has six carbon atoms in the basic unit.



Properties: 1. Nylon fibers are linear structures in which the molecular chains are arranged parallel to each other and held together by hydrogen bonding. The strong intermolecular hydrogen bonding makes the structure of these fibers crystalline and imparts them high strength, elasticity and melting point.

2. They are chemically stable and resistant to abrasion.

3. The nylon fabric is tough, lustrous, moisture resistant, easy to dye, retains color and can be heat on stream.

Applications: 1. Nylon is used for making gear, fishing lines and other recreational equipments.

2. It is used for making combs and bristles of the brushes.

3. It is used for making gears, bearings, bushes, etc.

4. Nylon6,6 is used for fabric for clothing, socks, sportswear, carpets, etc.

5. Nylon6 is used for making cords for tyres.

Silicone polymers

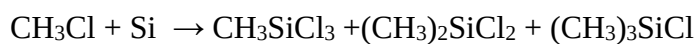
In silicone rubbers, the polymer chain does not involve long C-C chain, but contain alternate silicon and oxygen atoms in which Si atoms are joined to organic groups (methyl groups)

Preparation

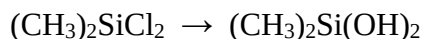
Preparation of silicone polymers involves three important steps.

Step1. Preparation of dimethyl silicone: It is obtained by reacting silicon with alkyl chloride or

reacting silicon tetra chloride with Grignard reagent

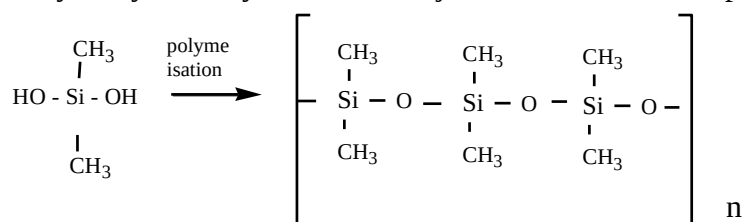


The dimethyl silicon chloride is separated by fractional distillation and hydrolysed to get dihydroxy dimethyl silicon



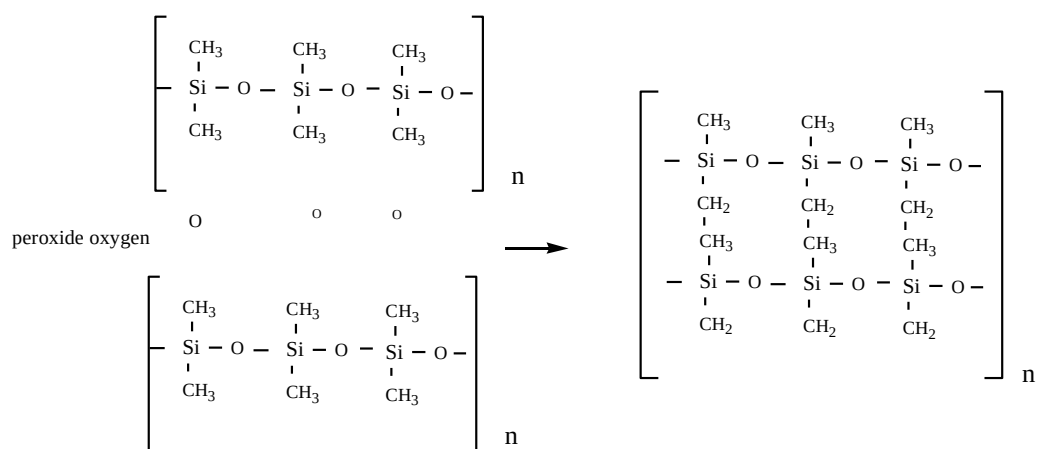
Step2; Polymerisation of dihydroxy dimethyl silicon

The dihydroxy dimethyl silicon is subjected to condensation polymerisation to get silicone polymer



Step3

Vulcanisation of silicone polymer: The silicone polymer is mixed with fillers like SiO_2 and vulcanized using the vulcanizing agent organic peroxides. During vulcanization dimethylene bridge is formed between methyl groups of adjacent chains.



Vulcanization increases hardness, rigidity and brittleness of silicone polymer.

Properties of silicone polymer

1. Silicone rubbers possess excellent thermal stability, resistance to prolonged exposure to sunlight, weathering, boiling water, dilute acids and alkalis.
2. They remain flexible in the temperature range of 90- 250°C..
3. They are nontoxic.
4. They are resistant to oxidation by air and ozone

5. Silicone resins have a very good heat resistance property and are good electrical insulators, particularly at elevated temperatures.

Applications : 1. Silicone rubber is used in making tyres for aircrafts since they prevent damage on landing due to their flexibility in the temp. range of 90-250°C.

2. It is used in making lubricants, paints, etc.

3. It is also used for making artificial heart valves.

4. Used in the manufacture of heat resistant glass cloth laminates, pressure sensitive adhesives, silicone rubbers and coatings.

Biodegradable plastics

They are degradable plastics in which degradation results from the action of naturally occurring microorganisms such as bacteria, fungi and algae. Biodegradable polymers can be classified as natural biodegradable polymers and synthetic biodegradable polymers.

Natural biodegradable polymers

They are produced in nature by all living organisms. Polysaccharides such as starch are the most prevalent natural biodegradable polymers in commercial use. Aliphatic polyesters such as polyhydroxy alkanoates (PHA) are also a family of easily biodegradable polymers found in nature.

In these polymers, the biodegradation reactions are typically enzyme-catalyzed and occur in aqueous media. Natural macromolecules containing hydrolysable linkages, such as proteins, cellulose and starch are generally susceptible to biodegradation by the hydrolytic enzymes of microorganisms.

Synthetic biodegradable polymers

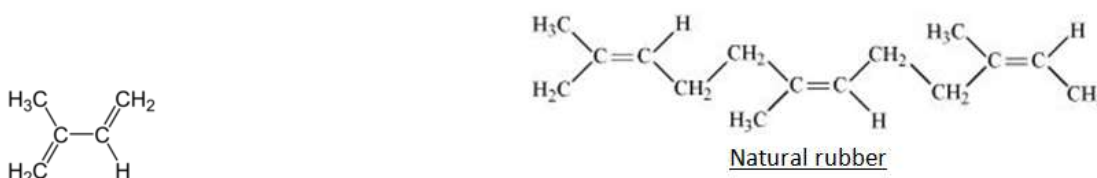
The major category of synthetic biodegradable polymers consists of aliphatic polyesters with a hydrolysable linkage along the polymer chain such as polylactic acid (PLA). Other widely available synthetic types include aliphatic/aromatic copolyesters. The biodegradation reactions of synthetic biodegradable polymers is same as that of natural polymers. These are enzyme-catalyzed and proceed in aqueous media.

Applications of biodegradable polymers

1. Food packaging.
2. Single-serve cups and other disposable foodservice items
3. Foam for industrial packaging, film wrapping, laminated paper, and in agricultural film products
4. Electrical appliances and electronics
5. Adhesives, paints and coatings.

Elastomers - Natural rubber

Natural rubber is found in several species of rubber trees, the most important being 'Hevia brasiliensis', and the rubber obtained from this is often called "Haeva rubber". Rubber is stored in the form of white fluid called latex behind the bark of the rubber tree. The latex oozes out when a cut is made on the bark. Natural rubber is a high polymer of isoprene (2-methyl 1,3-butadiene). The degree of polymerization is about 5000. Since poly isoprene contains C=C double bond, it exists in two geometrical isomer structures, cis and trans form. Natural rubber is soft and has a cis configuration.



Deficiencies of natural rubber

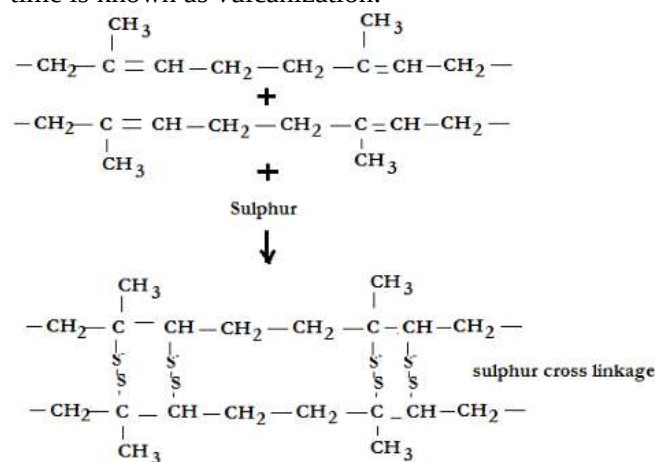
The virgin natural rubber as such has virtually no practical utility value because of its many inherent deficiencies mentioned below:

1. It is hard and brittle at low temperature, and soft and sticky at high temperatures.
2. It has poor keeping qualities as it develops foul odor on storage.
3. It is soft and soluble in many solvents. It has water absorption power.
4. It is a sticky substance, difficult to handle.
5. It is readily attacked by acids and alkalis. It has low tensile strength.
6. It has poor abrasion resistance.

To overcome the drawbacks discussed above, natural rubber is subjected to processes such as vulcanization and compounding to improve its properties.

Vulcanization

The process of heating and mixing the crude rubber with sulfur to a definite temperature for a specific time is known as vulcanization.



The sulfur combines chemically at the double bonds forming cross links between different polymer chains. This cross linking brings about stiffness in the rubber, and subsequently prevents intermolecular sliding of rubber polymer chains.

The extent of stiffness of vulcanized rubber depends on the amount of sulfur added. For example, a rubber tyre contains 3-5% of sulfur, whereas a battery case made from

rubber may contain as much as 30% of sulfur. Apart from sulfur, other vulcanizing agents like hydrogen sulfide, sulfur monochloride, benzoyl chloride, etc.

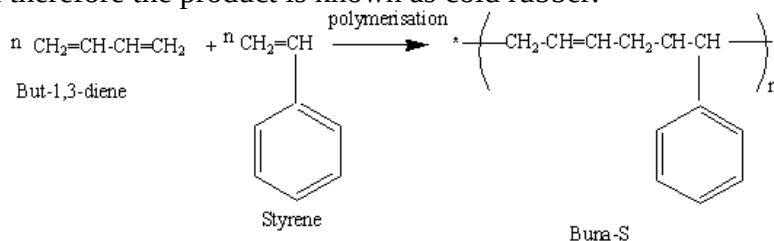
Properties of vulcanized rubber

1. It has good tensile strength.
2. It possess low water absorption capacity.
3. It has higher resistance to oxidation and abrasion, wear and tear resistance.
4. It is resistant to organic solvents such as petrol, benzene, CCl₄ and fats and oils..
5. It exhibits good electrical insulation property.
6. It has a wide useful temperature from – 40 to 100°C

Synthetic rubbers

1. Buna-S (SBR or GRS)

It is a copolymer of styrene and butadiene. It is called SBR- styrene- butadiene rubber and GRS – Government controlled styrene rubber. It is manufactured by polymerization of 75 parts of butadiene and 25 parts of styrene in presence of peroxide initiators. The polymerization is carried out at 5°C and therefore the product is known as cold rubber.



Properties

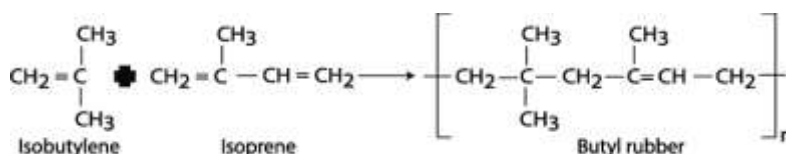
Since SBR contains double bonds, vulcanization can be done in the same way as natural rubber using sulfur or sulfur monochloride. Vulcanized Buna-S has high abrasion resistance and good load-bearing capacity. It can be easily attacked by oxidizing agents such as oxygen and ozone.

Applications

SBR is used in the following applications : a) Motor tyres; b) shoe soles; c) insulation of wires and cables; d) Gaskets; e) Adhesives and f) tank-lining

Butyl Rubber

It is a copolymer of isobutylene and a small amount of isoprene. It is produced by the cationic low temperature (- 90°C) copolymerization of isobutylene(90%) in the presence of a small amount(10%) of isoprene. The use of isoprene is to provide necessary residual unsaturation for vulcanization.



Properties

1. Under normal conditions it is amorphous, but it crystallizes on stretching.

- ## Applications

- ## Neoprene

$$n \text{ H}_2\text{C}=\text{CH}-\text{C}(\text{Cl})=\text{CH}_2 \longrightarrow \left[\text{H}_2\text{C}-\text{CH}=\text{C}(\text{Cl})-\text{CH}_2 \right]_n$$

2-chlorobuta-1,3-diene (chloroprene) poly(2-chlorobuta-1,3-diene) (polychloroprene or 'neoprene')

Properties

- ## Applications

1. Neoprene rubber can replace natural rubber in most of the applications. It is used for hoses, tubes for carrying oils and chemicals, gloves, cable and wire coatings, belts, shoe heels, tyres, etc.

LUBRICANTS

Lubricant is a substance used in between two moving surfaces to reduce the friction. Lubrication is a process of reducing friction and wear between two surfaces by adding lubricant in between them.

Characteristics of a lubricant

1. A good lubricant should not undergo any decomposition, oxidation or reduction at high temperature.
2. It should have higher flash and fire points than the operating temperatures.
3. A good lubricant should have high oiliness, viscosity index and aniline point.
4. It should not corrode the machine parts.

Classification of lubricants

1. Liquid lubricants
2. Semi solid lubricants
3. Solid lubricants
4. Emulsions

Liquid lubricants or Lube oils

I. Vegetable oils: They are glycerides of higher fatty acids and have very good oiliness. But they have the following limitations:

- i) They undergo oxidation at higher temperatures.
- ii) They are hydrolyzed easily under moist conditions

Vegetable oils are actively used as blending agents with other lubricating oils.

II. Mineral oils or Petroleum oils

They are obtained by fractional distillation of crude petroleum oil. The length of hydrocarbon chains varies from C_{12} to C_{50} . It is cheap and stable under normal operating conditions. It possesses poor oiliness. It can be improved by mixing it with animal or vegetable oils.

III. Synthetic lubricants

Examples of synthetic lubricants are silicones, poly glycols, ethers, etc., which have been developed for effective use in the temperature range of -50 to 250°C .

Semi solid lubricants (Greases)

Greases are semisolid lubricants obtained by thickening of lubricating oils by the addition of metallic soaps. Usually Na, Ca and Li soaps are used. eg. Lithium grease

Solid lubricants

The two most usual solid lubricants employed are graphite and molybdenum disulfide. They are used either in the dry powder form or mixed with water or oil.

I. Graphite: Graphite is the widely used of all solid lubricants. It is very soapy in nature, non-inflammable and is not oxidized in air upto 375°C . In the absence of air, it can be used upto higher temperatures. Graphite is used either in powdered form or as a suspension. Suspension of graphite in water or oil is prepared with the help of an emulsifying agent like tannin. When graphite is dispersed in water it is called aqua dag and, when it is dispersed in oil it is called oil dag. Oil dag is found useful in internal combustion engines. Water dag is useful where a lubricant free from oil is needed. Eg. in foodstuff industry.

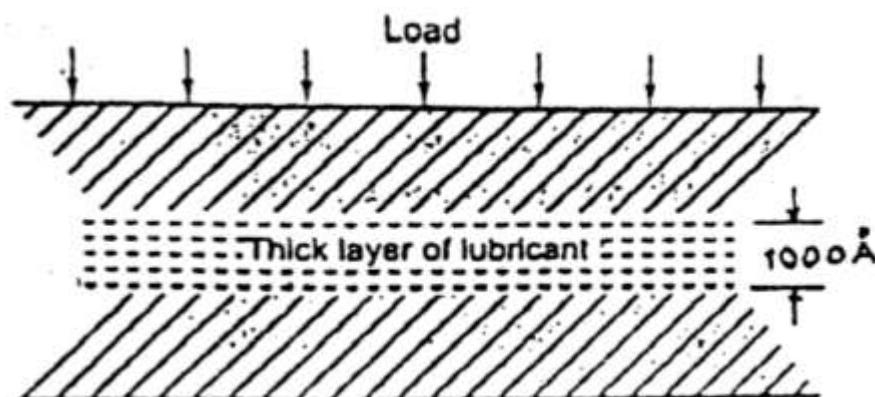
II. Molybdenum disulfide: It is stable in air upto 400°C . It has a sandwich like structure in which a layer of molybdenum atom lies between two layers of sulfur atoms. The atoms in the layers are bonded together by strong covalent bonds but the layers are held together by weak Vander Waal forces of attraction.

The molybdenum and sulfur layers slide over one another. It is used in heavy machineries working under heavy load and high temperature.

Mechanism of Lubrication

a) Thick Film or Fluid Film or Hydrodynamic Lubrication

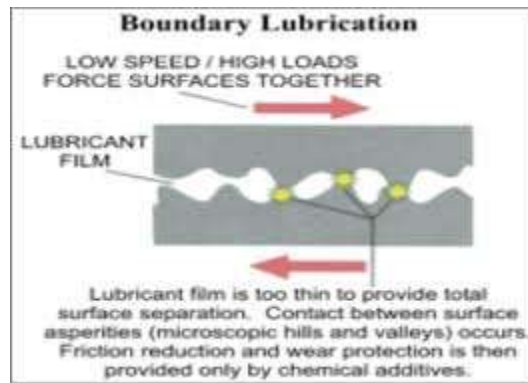
It is carried out with the help of liquid lubricants. In this mechanism, two moving and sliding surfaces are separated by thick film of lubricant fluid of about $1000\text{\AA}$, applied to prevent direct surface to surface contact and consequently reduce wearing and tearing of metals. Therefore, it is known as thick film or fluid film lubrication or hydrodynamic (hydro meaning liquid and dynamic meaning relative motion) lubrication.



In this case fluid is formed by mixing of hydrocarbon oils and anti-oxidants with long chain polymer so as to maintain viscosity. Fluid film lubrication is useful in delicate and light machines like watches, clocks, guns, scientific equipments.

b) Thin Film or Boundary Lubrication

It is carried out with semi-solid (grease) and solid (graphite and molybdenum disulphide) lubricants. Boundary lubrication is a condition in which the lubricant film becomes too thin to provide total separation. In this type of lubrication a thin film of lubricant is adsorbed on the surface by weak Vander Waals forces.



Thin film lubrication is operating at relatively low speed and heavy loading (pressure).

c) Extreme Pressure (or Temperature) Lubrication

In this mechanism, moving or sliding surfaces are under high pressure and speed, therefore this is known as extreme pressure lubrication. Under these condition liquid lubricants may fail to stick and decompose or vaporize. These problems are minimized by adding special additives to mineral oils. These additives form durable films on metal surfaces which can withstand high loads and high temperatures. Important additives are organic compound having group like chloride, sulfur, phosphorus etc. They react with metallic surface to form metallic compound (possess high melting points and serve as good lubricants under extreme temperatures and pressures) like chlorides, sulphides, phosphate as more durable film⁴, sulfides, phosphate as more durable film

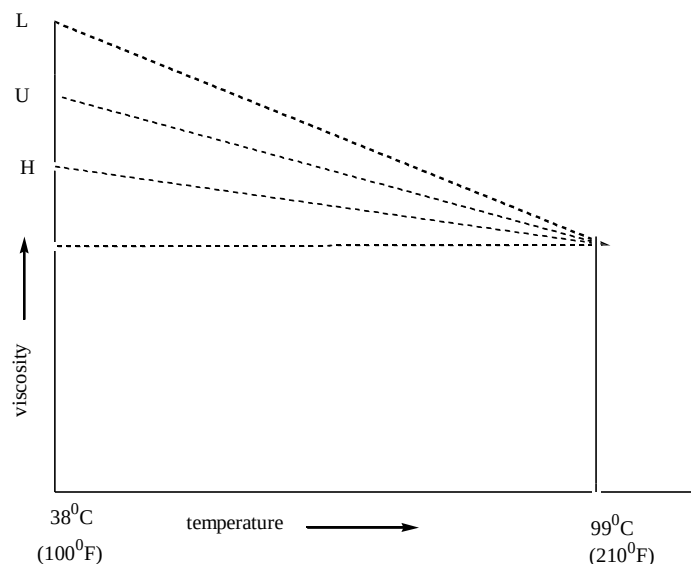
Properties of lubricants

Viscosity: Viscosity is a measure of internal resistance of a liquid during its flow. It is expressed in centi poise. The viscosity of an oil is the time in seconds for a given quantity of oil to pass through a standard orifice under the specified conditions.

Viscosity Index: The viscosity of an oil decreases with increase in temperature. The rate of change of viscosity with temperature is indicated by viscosity temperature curves by a scale known as Viscosity index (V.I.). V.I. is a measure of rate at which the viscosity of an oil changes with temperature.

For a good lubricating oil, variation of viscosity with temperature should be small. i.e., V.I. is high. For the measurement of viscosity index, two types of standard oils are used. (1) Paraffin based Pennsylvanian oils (V.I.100, designated as 'H' oils) and (2) Naphthenic based gulf oils(V.I. 0, designated as 'L' oils). The viscosity of the oil under test is measured at 100°F and also at 210°F. Let these values be 'U' and 'V' respectively. The difference between the two values should be low if the oil is good. From the list of 'H' oils, the oil which has the same viscosity at 210°F is selected and its viscosity is measured at 100°F. Let it be 'H'. Then from the list of 'L' oils, the oil which has the same viscosity at 210°F is selected and its viscosity is measured at 100°F and let it be 'L'. The viscosity index of the oil under test is given by

$$V.I = \frac{L-U}{L-H} \times 100$$



V.I. of an oil can be increased by the addition of linear polymers such as poly isobutylene ,n-hexanol etc.

Flash point and fire point

Flash point is the lowest temperature at which the oil gives off enough vapors that ignite for a moment, when a small flame is brought near it.

Fire point is the lowest temperature at which the vapor of the oil burns continuously for at least 5 seconds, when a small flame is brought near it. Generally, fire point is 5-40°C higher than flash point.

A knowledge of flash and fire points is useful in providing protection against fire hazard during transport and storage. A good lubricating oil should have flash and fire points higher than the operating temperatures of the machine.

Cloud and pour point

Cloud point of an oil is the lowest temperature at which the oil becomes cloudy when it is cooled slowly. Pour point is the temperature at which the oil ceases to flow or pour.

Most of the petroleum based lubricating oils contain dissolved paraffin wax and asphaltic impurities. When the oil is cooled, these impurities undergo solidification which causes jamming of the machine. A good lubricant must have low cloud point and pour point. Pour point of lubricating oil can be lowered by dewaxing and adding a pour point depressant.

Aniline point

It is the lowest temperature at which the oil is completely miscible with an equal volume of aniline. It gives an indication of the deterioration of an oil when it comes in contact with rubber sealings, packings, etc. Generally aromatic hydrocarbons have a tendency to dissolve natural or synthetic rubbers. A higher aniline point indicates lower percentage of aromatic hydrocarbons and hence less solvent effect on rubber packings. So, a higher aniline point is desirable.

Refractories

Materials that can withstand very high temperatures, over 3000°C , without degradation, softening or deformation are called refractories. These includes ceramics and superalloys, and are used in heat insulation of furnaces, jet and rocket engines, equipments in metallurgy, engineering and chemical industries. They are resistant not only to abrasive and corrosive action of molten metal, slag and gases at high operating temperatures, but also loss of heat and thus help in maintaining the high temperature conditions required for these processes.

Classification of refractories

Refractories are classified into three main types, based on the chemical properties of their constituent substances:

1. Acid refractories: These are made up of acidic materials such as alumina (Al_2O_3) and silica (SiO_2). They are not attacked by acidic materials, but easily attacked by basic materials. Examples of acidic refractories are alumina, silica and fire-clay refractories.
2. Basic refractories: These are made up of basic materials such as CaO , MgO , etc.. These are not attacked by basic materials, but can be attacked by acidic materials. Examples of basic refractories are magnesite and dolomite refractories.
3. Neutral refractories: These are made from weakly acidic or basic materials such as carbon, Chromite ($\text{FeO} \cdot \text{CrO}_2$), zirconia (ZrO_2), etc. Examples of neutral refractories are, graphite, chromite, zirconia and carborundum (SiC) refractories

Properties of refractories

1. Refractoriness: It is the ability of the refractory material to withstand heat without appreciable softening or deformation under given set of conditions. Most of the commercial refractories do not exhibit sharp melting point and they soften gradually over a range of

temperatures. This is due to the fact that they are composed of several minerals. Therefore, the refractoriness is generally measured as the softening temperature. It is necessary that a refractory material should have a softening temperature higher than the operating temperature of the furnace in which it is to be used.

The softening temperatures of refractories are generally determined pyrometric cone (or seger cone) test. In this test, behavior of heat on cone of refractory (to be tested) and series of pyrometric cones of standard dimensions are compared. These cones are small pyramid shaped with triangular base. The test refractory in the form of a cone is kept alongside similar sized standard cones and all are heated uniformly. As the seger (pyrometric) cones are made of a particular refractory of a definite softening temperature, they are assigned ascending seger cone numbers with increasing softening temperature. When the test cone softens and loses its shape, one of the standard seger cones also softens and loses its shape provided its refractoriness is close to that of the test cone. The serial number of this standard seger cone is noted and this number is the pyrometric cone equivalent (PCE) of the refractory under that test. When the test cone softens earlier than one standard cone, but later than the previous one, the PCE value of the test sample is approximated as the average of the two.

2.Refractoriness – under – load (RUL) (or Strength): Temperature resistance and load bearing capacity are two essential qualities of a refractory. This is due to the fact that the commercial refractories which are used for lining high temperature furnaces are expected to withstand varying loads of the charge. Hence, they should possess high strength and excellent temperature resistance. The mechanical strength of refractories is expressed in terms of refractoriness-under-load (RUL)

RUL test is performed by applying a constant load to the refractory specimen and heating in a carbon resistant furnace at a standard rate of 10°C per minute. The deformation of the test piece is measured simultaneously till it undergoes 10% deformation. The RUL is expressed as the temperature at which 10% deformation takes place. The RUL of a high temperature refractory under a load of 3.5kg/cm^2 is about 1350°C ; for a moderate temperature refractory, it is 1100°C .

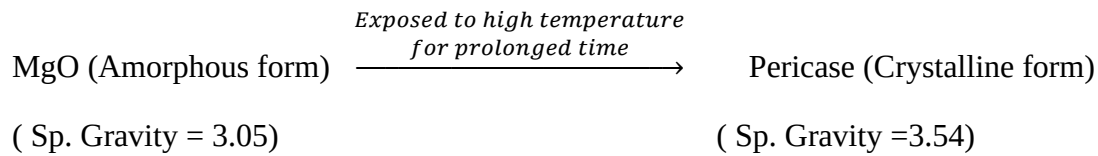
3. Dimensional stability: It is the resistance of a material to any change in volume when it is exposed to high temperature, over a prolonged time.

The dimensional changes can be permanent contraction or permanent expansion.

a) Permanent contraction: It occurs if a refractory material transforms from one crystalline form to another of lower density when exposed to high temperature. For instance, silica exhibits such crystalline changes as follows:

Quartzite	Tridymite	cristobalite
Crystalline form of silica	Orthorhombic form of silica	Cubic form of silica
(Sp. Gravity=2.65)	(Sp. Gravity=2.26)	(Sp. Gravity=2.32)

b) Permanent expansion: It occurs when a refractory is exposed to high operating temperatures for a prolonged period of time. Such expansion in volume of refractory material affects the transformation of crystalline forms into one another of higher densities. For instance, magnesium oxide may transform to a denser crystalline form as follows:



4. Porosity: All refractories possess pores, openings or minute channels that are formed during the manufacturing process. The number or extent of refractory pores is an important parameter as it has direct influence on its mechanical strength, chemical stability, and abrasion resistance. If a refractory material has greater porosity, gases and sludge may enter into the crevices and adversely affect the life of the material. Thus, a refractory material with a low porosity is an ideal situation that can resist corrosion and shock

5. Chemical inertness: A good refractory material must necessarily possess chemical inertness and should not form fusible compounds with ash content, furnace gases, sludges and slag materials.

6. Thermal expansion: An ideal furnace design allows for thermal expansion of refractory as all solids expand on heating and contract on cooling. A good refractory material must possess least thermal expansion, since it adversely affects furnace design, and repeated expansion and contraction may cause wear of the refractory lining and overall furnace structure, making it prone to structure failure.

7. Thermal spalling: Cracking, peeling, or fracturing of a refractory material under high operating temperature is called thermal spalling. A good refractory must possess excellent resistance towards spalling. Spalling can be reduced by avoiding sudden furnace temperature changes and protecting the refractory from undue stress.

Cement

An important engineering material, generally employed in construction, cement is known to have adhesive and cohesive properties that can bond materials such as bricks and stones. Cements are of 4 types: natural, Puzzolana, Slag and Portland.

Portland cement

Portland cement is a completely pulverized finely ground powder obtained by calcining a known proportion of a mixture of argillaceous and calcareous raw materials together at 1500°C.

Manufacture of Portland cement

The following are the raw materials required for the manufacture of Portland cement.

- a) Calcareous materials such as lime, limestone, chalk, etc., having calcium oxide.
- b) Argillaceous materials such as clay and slate having alumina and silicon dioxide.
- c) Pulverized coal and gypsum.

Manufacture of Portland cement involves the following three processes – mixing, burning and grinding.

Mixing of raw materials

Mixing of raw materials is performed either by dry or wet process.

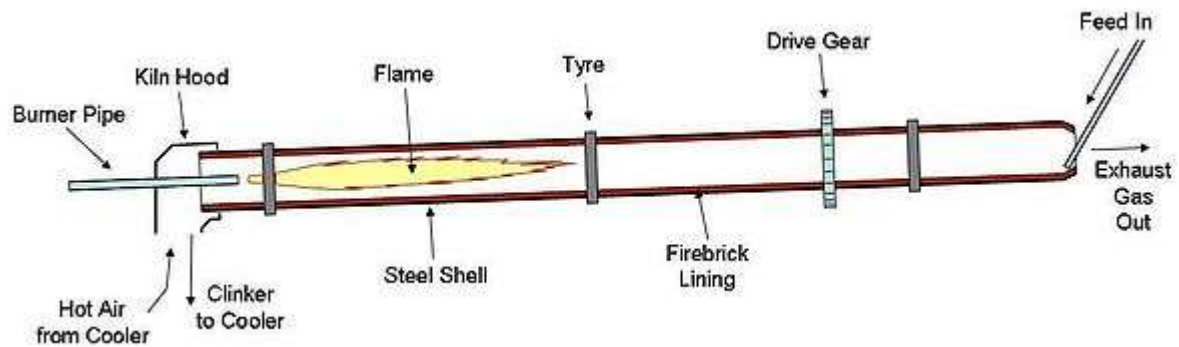
Dry mixing process: In this process, limestone (or chalk) and clay are crushed separately to obtain pieces of 2-5cm size. The pulverized powder is further ground to fine powder and the powdered limestone and clay are placed in separate hoppers. These pulverized materials are mixed in requisite proportions to obtain a 'dry-mix'. The proportion of raw materials used in the dry process is as follows:

- a) Lime (CaO)- It is the principal constituent of cement (60 – 69%) and its proportion is regulated because if there is an excess or lack of lime, it reduces the strength of cement, causing disintegration.
- b) Silica (SiO_2): It is an important constituent (17 – 25%) that imparts strength to cement
- c) Alumina (Al_2O_3): It constitutes 3-8% of cement and imparts strength to allow quick setting. An excess of alumina can weaken the cement bonding
- d) Iron oxide (Fe_2O_3): It constitutes about 2-4% and imparts strength, color, and hardness to the cement.
- e) Magnesium oxide (MgO): It constitutes about 1-5% and any excess of it cause disintegration of cement.
- f) Sulfur trioxide (SO_3): Sulfur trioxide, in trace amounts, is desirable as it imparts soundness of cement
- g) Alkali oxides: ($\text{Na}_2\text{O} + \text{K}_2\text{O}$): Constitutes only 0.3-1.5%. An excess of alkali results in effervescence of cement and may cause disintegration.

Wet mixing Process:

During the wet process, calcareous raw materials are powdered and stored. Clay is thoroughly mixed with water to eliminate any organic matter and stored separately. Both these raw materials are then mixed in calculated proportions and sent to grinding mills, where about 40% water is added to form the slurry and is sent to the rotary kiln.

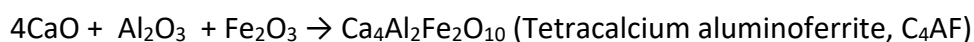
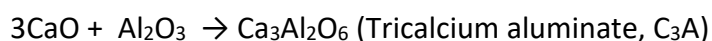
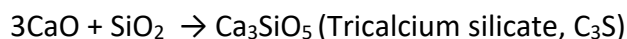
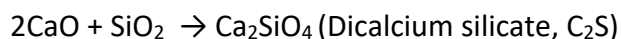
Burning



The rotary kiln is a long steel tube of about 90 – 120 m and about 2.5 – 3m in diameter. The inner lining of the kiln is made of refractory bricks. Rotary kiln is kept in an inclined position of 1 in 25 to 1 in 30, which is allowed to rest on roller bearings. The cement kiln is allowed to rotate at 1rpm about its longitudinal axis. Coal and air are allowed to pass from the lower end of the kiln for obtaining long flame that heats up the inner lining to about 1750°C.

During the manufacture of cement in rotary kiln, the slurry is passed through the upper part of the heated kiln called the drying zone. Due to the high temperature (above 400°C) most of the water in the slurry evaporates. The dried slurry glides down towards the hotter central zone (1000°C) called the calcination zone, where it decomposes forming quicklime (CaO) and carbon dioxide. The CO₂ escapes leaving behind a lumpy mass, which reaches the lower portion of the kiln, clinkering zone, maintained at 1750°C where clay and lime undergo fusion resulting in the formation of calcium aluminates and silicates, that is cement.

The chemical reactions are represented as:

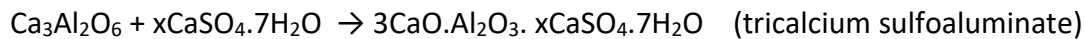


The aluminates and silicates fuse together forming small clinkers which are of about 1cm diameter stones and hard, grey in color.

Grinding

The cooled clinkers are ground in ball mills with about 2 – 3% powdered gypsum (CaSO₄), to prevent early setting of cement. Gypsum retards the dissolution of tricalcium aluminate by

forming tricalcium sulfoaluminate, which is practically insoluble and prevents early setting and hardening of cement.



Setting and hardening of cement

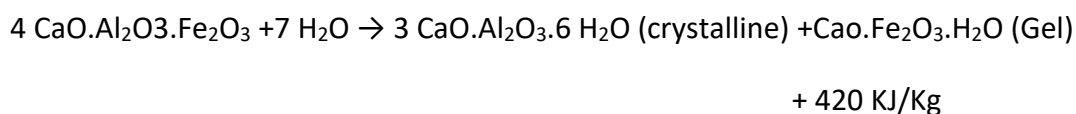
When water is added to cement, it first forms a paste which then hardens; the process is referred to as setting and hardening of cement. Setting involves initial stiffening of cement paste due to gel formation, whereas hardening refers to strength development due to crystallization. Thus, the extent of gelation followed by crystallization dictates the strength of the cement.

Reactions involved in setting and hardening of cement: -

When cement is mixed with water, the paste becomes rigid within a short time which is known as initial setting. This is due to the hydration of tricalcium aluminate and gel formation of tetra calcium aluminoferrite.



tricalcium aluminate hydrated tricalcium aluminate (crystalline)

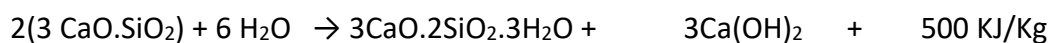


Dicalcium silicate also hydrolyses to tobermonite gel which contributes to initial setting



Dicalcium silicate Tobermonite gel

Final setting and hardening of cement paste is due to the formation of tobermonite gel and crystallization of calcium hydroxide and hydrated tricalcium aluminate.



tricalcium silicate Tobermonite gel calcium hydroxide
(crystalline)

During setting and hardening of cement, some amount of heat is liberated due to hydration and hydrolysis reactions.

After initial hydration of tricalcium aluminate and tetra calcium aluminoferrite, hydration of tricalcium silicate is carried out within 24 hours and completed in 7 days. Further, the aluminate gel crystallizes out simultaneously and dicalcium silicate begins to hydrate that

completes in 28 days. In a nutshell, we can say that strength enhancing takes place between 7 to 28 days due to hydration of dicalcium and tricalcium silicates.